

HARRISON JANSMA

Senior Applied Scientist · Generative AI & Applied ML Solutions

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SUMMARY

Senior applied scientist / ML engineer with 6+ years taking applied AI from research to production for enterprise customers — generative AI, real-time NLP, and large-scale ML systems with measurable business impact. Specialized in agentic LLM workflows, retrieval-augmented generation, low-latency serving, and rigorous evaluation. Proven record translating ambiguous business problems into shipped models serving thousands of concurrent users and leading cross-functional teams.

CORE SKILLS

Applied Science & GenAI LLM orchestration, agentic / multi-step reasoning workflows, prompt engineering, retrieval-augmented generation (RAG), fine-tuning, transformers (DistilBERT/BERT), embeddings, NLU, sentiment & emotion classification, reinforcement learning, simulation

ML at Scale PyTorch, TensorFlow, Hugging Face, scikit-learn; streaming inference (Apache Kafka, AWS Lambda, DynamoDB); vector databases; low-latency serving; AWS (S3), Snowflake

Evaluation & Observability LLM-as-a-judge, offline & online evaluation, quality-metric design, governance dashboards, New Relic, PagerDuty, Splunk

Business Impact Credit-decisioning models, agentic GenAI, customer-facing solution delivery

Languages & Data Python, SQL, PySpark, Hive, Tableau

EXPERIENCE

Capital One — Plano, TX

2020 – Present

Data Science Manager (2024–Present) · Principal Data Scientist (2022–2024) · Sr. Associate, Data Science (2021–2022) · Data Process Manager (2020–2021)

Applied Generative AI (Current Focus)

- Lead Generative AI for Capital One's contact centers — applied LLM / agentic systems that augment thousands of servicing agents live, mid-conversation, with measurable business outcomes.
- Built an agentic GPT reasoning pipeline (Kafka) that auto-drafts agent notes, cutting after-call work with projected operational savings at full rollout; added an LLM "agent-as-a-judge" review stage before drafts reach the agent.
- Designed a retrieval-augmented generation system surfacing the right documents mid-task via vector-embedding retrieval over live transcripts (Kafka, AWS Lambda, vector database) — in real time.
- Shipped an LLM complaint-detection system on live transcripts, sharply improving capture over the legacy ML classifier at high precision.

Applied ML at Scale

- Architected Capital One's first real-time AI system for detecting frustrated customers, monitoring live calls at scale and alerting in real time.
- Engineered parallel real-time inference pipelines with high throughput at low latency (Kafka, DynamoDB, S3, Snowflake).
- Trained and deployed DistilBERT and LLM models (PyTorch, Hugging Face, scikit-learn) for streaming sentiment and NLU.

Applied Science in Credit & Leadership

- Engineered Capital One's auto-loan underwriting ML suite, decisioning loans at large scale; built and owned the core loan-decisioning model; built a simulation engine for RL-based credit-policy optimization.
- Lead a cross-functional team of ML engineers and data scientists, setting technical objectives with VP- and EVP-level stakeholders.

Sprint — Irving, TX

2019

Data Science Intern

- Designed a model-training framework for anomaly detection on customer billing data — 5x increase in detectable failure types; only extension across an 80+ intern class.

University of Pennsylvania (via Upwork) — Remote

2018 – 2019

Machine Learning Researcher — NLP

- Built an emotion-classification model and processed millions of public social-media profiles for a population-level NLP study.

EDUCATION

- **M.S. Computer Science (Data Science)** — The University of Texas at Dallas · 2018–2020
- **B.B.A. Mathematics, Business Fellows** — Baylor University · 2013–2017 — Magna Cum Laude, GPA 3.86; Honors Program